Predicting Cyanobacteria Blooms in Peri-Alpine Lakes

A machine learning approach trained on Lake Greifensee and applied to Lake Neuchâtel

Diego Segura
University of Neuchâtel
June 2026

Contents

Abstract

Cyanobacteria are a massive problem in the Lake Neuchâtel for some times. For this reason, we decided to train a model to predict the probability of a cyanobacteria bloom during the year from a rich dataset made by scientists on the Lake Greifensee.

The model shows us a higher probability of blooms during autumn through early spring, and a lower probability in summer and deep winter.

1. Introduction

1.1 Context

Cyanobacteria (*Cyanobacteriota*) are a microscopic group of photosynthetic bacteria found naturally in fresh waters worldwide. Cyanobacteria can rapidly and massively multiply and form enormous population explosions called “blooms” when the conditions are auspicious to the increase, especially when the water has the right temperature and full of rich nutriments.

An exposure to these toxins or consuming contaminated water can cause serious illness in humans, pets, and wildlife.

1.2 Objective

The research has been made to estimate the probability of a cyanobacteria bloom from environmental conditions, and produce a monthly risk curve for Lake Neuchâtel.

2. Data

2.1 Source

The machine learning model has been trained on a dataset from the EAWAG automated platform.

This dataset contains many really interesting features that can give us a good base to make a model train on the dataset. For example, there is environmental features like the mean temperature of the water, the Schmidt stability of the lake or even the concentration of various nutriments.

At first, the model was supposed to be trained on US lakes but was dropped because cross-lake sampling bias hid the temperature signal, which is why we switched to a single, well-monitored lake.

2.2 Target variable

The target is a binary (2-class) variable made on the mean of the phycocyanin, the main pigment of the cyanobacteria. A “bloom” day is defined when the phycocyanin is above its 75th percentile and when under, it is defined as a non-bloom day.

The phycocyanin values in this dataset are in sensor-specific relative units, not calibrated to standard measures like cells per millilitre. Without that calibration there is no WHO or regulatory threshold that can be applied directly to these numbers. Using the 75th percentile is a principled, data-driven alternative: it flags days when cyanobacteria levels are in the upper quarter of what Lake Greifensee showed over the monitoring period, which serves as a working definition of “elevated” for a lake without an established absolute baseline. It also keeps the class balance at around 25% bloom days, which would not be guaranteed with an arbitrary fixed value.

2.3 Features

Since the dataset contains several features, we needed to select which ones were useful for the research and the model to be trained on and which ones were either out of context, either were set aside to avoid data leakage.

For the analysis of the Lake Greifensee, we selected the following features: mean_temp (mean water temperature), mean_schmidt (mean of the Schmidt stability), mean_par (mean of the photosynthetically active radiation), windspeed_mean (windspeed mean), precipitation_tot (total of the precipitation), month_sin (calculated from the date), month_cos (calculated from the date).

For probalility of bloom prediction in the Lake Neuchâtel, we focused more on environmental features that can be transposed between both lakes: temperature_2m_mean (from Open-Meteo), wind_speed_10m_max (from Open-Meteo), precipitation_sum (from Open-Meteo), shortwave_radiation_sum (from Open-Meteo), sunshine_duration (from Open-Meteo), month_sin (calculated from the date), month_cos (calculated from the date).

3. Methodology

3.1 Preprocessing

As it was said earlier, before being able to use the dataset as it owns, we needed to do a cleanup and generate extra variables, features to have a dataset that matches our requirements.

Since the mean of phycocyanin was our main variable (the target), we removed every observation where the value of the phycocyanin variable were omitted. To have a complete year cycle, the season were encoded as sine/cosine, that way January was close to December mathematically speaking. The last features manually encoded were every feature related to the environmental condition as the air temperature or the wind speed. To get this data we used the Open-Meteo API and coordinates of the two Swiss lakes.

3.2 Models

Logistic Regression was used as an interpretable baseline, the way we can see if the project has a predictive capacity or not. We trained this model on scaled data since it ensures smooth gradient descent convergence and keeps penalties equal across all features.

For the main predictive model, we used a Random Forest, because it seems that the relation between features and target may not be linear. The trees from the random forest are inherently scale-invariant, meaning it performs identically whether your data ranges from 0 to 1 or from 0 to 1’000’000.

3.3 Evaluation

Performance was estimated with stratified 5-fold cross-validation. A single temporal hold-out was not practical here: with only around 18 months of data and a strong seasonal cycle, any contiguous test block tends to cover just one season and contain no bloom days, making evaluation meaningless. Stratified sampling ensures each fold preserves the ~25% bloom rate.

Two metrics are reported: ROC-AUC measures how well the model separates bloom from non-bloom days, while PR-AUC is more relevant given the class imbalance. Balanced class weights were applied during training so the model does not simply default to predicting the majority class.

4. Results (Lake Greifensee)

4.1 Model performance

The Random Forest model was kept as the main predictive model. To understand this decision the performance of both models needs to be studied.

4.1.1 Full features

Logistic Regression, full features ROC-AUC : 0.68 ± 0.03

ROC-AUC is 0.68. Since 0.5 means the model has no predictive capacities, it is still an honest result but not the best neither.

Random Forest, full features ROC-AUC : 0.902 ± 0.029

Random Forest, full features PR-AUC : 0.715 ± 0.062

ROC-AUC is 0.90 and PR-AUC is 0.72, well above the 0.25 base rate, so the model predicts blooms quite well. The cross-validation keeps this honest, since each fold is tested on data it wasn't trained on.

4.1.2 Full features

Logistic Regression, transposable features ROC-AUC : 0.651 ± 0.082

ROC-AUC is 0.65. Since we removed features related to the Lake Greifensee, we lost few performances on the model but we can still see that the trend stays quite the same as the full features model.

Random Forest, transposable features ROC-AUC : 0.819 ± 0.035

Random Forest, transposable features PR-AUC : 0.576 ± 0.07

ROC-AUC is 0.82. Same as the full features model, the Random Forest has better results than the Logistic Regression. We also lose few performances from the full features model but it is still pretty good for a transposable model.

4.2 Interpretation

The interpretation will be done on the full features model, since it is richer and more efficient.

mean_schmidt	0.224862
mean_temp	0.220569
mean_par	0.186943
month_cos	0.123210
windspeed_mean	0.101968
month_sin	0.087993
precipitation_tot	0.054456

Table 1 Importance of features

This table represent the importance of every feature in the model. Schmidt is the most used variable, then water temperature and light. The importances are fairly spread out, so the model isn't just reading off the calendar month, it really uses the environmental conditions.

The usual picture of a cyanobacteria bloom is a green scum on a warm lake in August, and for shallow, nutrient-rich lakes that is accurate. In deep peri-alpine lakes like Greifensee and Neuchâtel, the dominant species is Planktothrix rubescens, which behaves very differently. During summer, strong thermal stratification keeps it confined to a cooler intermediate layer well below the surface, where it stays largely invisible to sensors measuring the upper water column. When temperatures drop in autumn and stratification breaks down, the water column mixes and brings Planktothrix up into the upper layers, where the phycocyanin sensor captures it. The result is a bloom signal that peaks in cold, mixed, low-light conditions rather than in warm summer; the opposite of what most people would expect.

4.3 Seasonal pattern

We generated graphs to understand which period is more susceptible to an important bloom of cyanobacteria and how the blooms evolve during the calendar year.

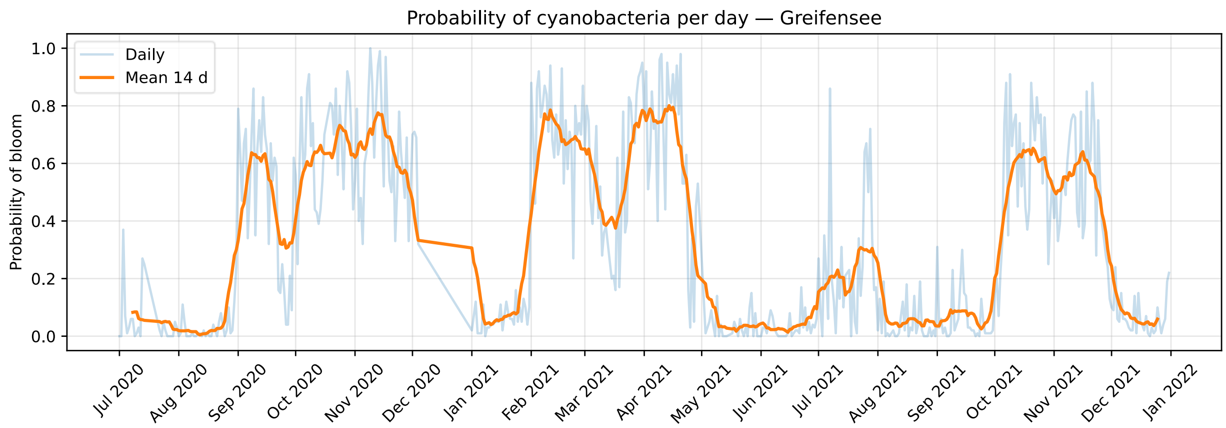

Figure 1 Cyanobacteria per day, Greifensee

The seasonal cycles show peaks in spring and autumn, lows in summer and deep winter.

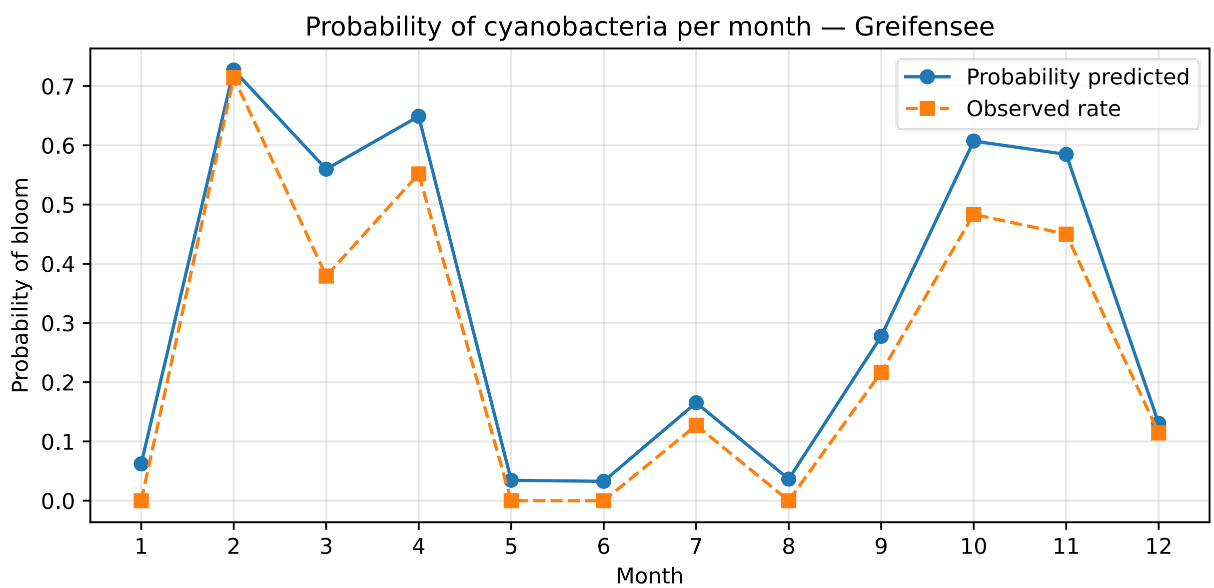

Figure 2 Comparison between predicted and actual values

The two curves follow each other closely, so the model reproduces the seasonal, day by day, pattern well. The predicted line sits a bit above the observed one because the balanced class weights push the probabilities up, so read the shape here rather than the exact level.

5. Application to Lake Neuchâtel

As it was mentioned earlier, the transposable model mainly remains on weather and season variable, due to a lack of open data in Lake Neuchâtel. However, the probability curve still follows the same trend as the full features model and the same peaks and lows.

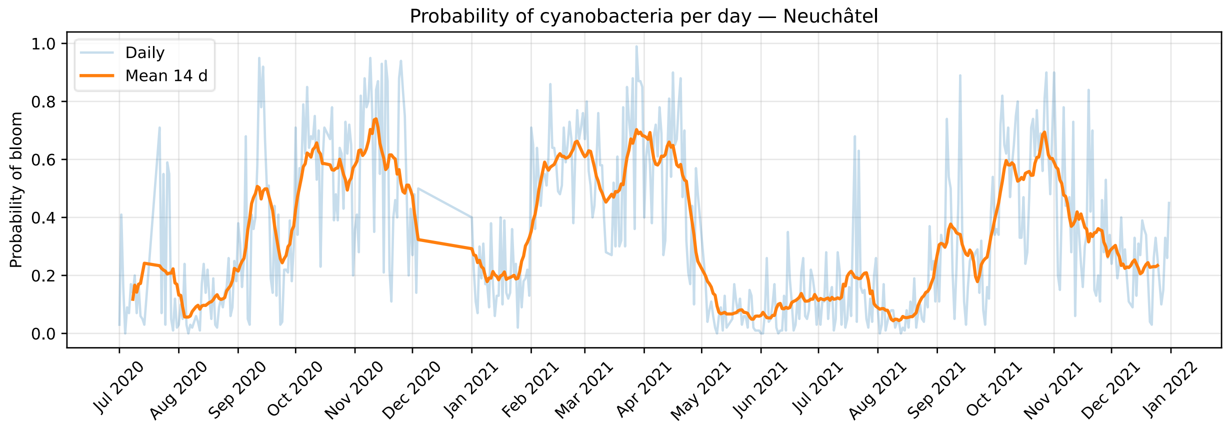

Figure 3 Cyanobacteria per day, Neuchâtel

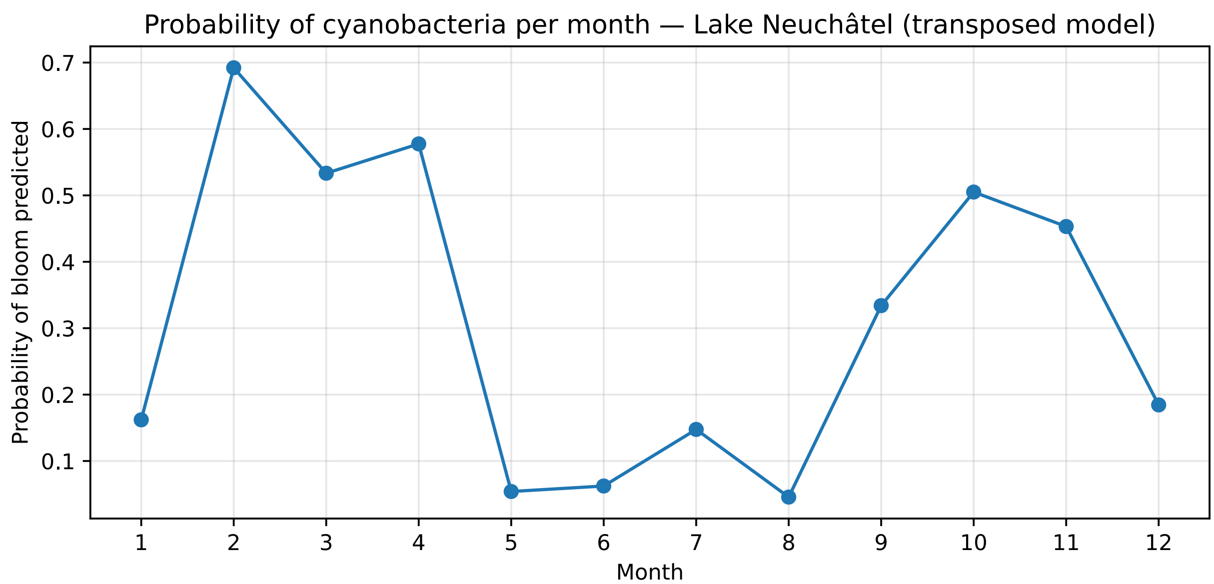

Figure 4 Cyanobacteria per month, Neuchâtel

These graphs show us a real similarity between the Lake Greifensee and the Lake Neuchâtel. However, this is more an extrapolation by analogy between two similar peri-alpine lakes because it was not validated on real Neuchâtel measurements.

6. Discussion

6.1 Main findings

The model tends to say that blooms are most likely happening during autumn through early spring. This seems at first counterintuitive, yet, the mean of every variable clearly shows that it has to happen like that.

	mean_temp	mean_schmidt	mean_par	windspeed_mean
target				
0	14.144389	556.072294	9.152513	1.946966
1	11.207590	297.144333	4.245481	1.703937

Table 2 Mean of variables

This table present the mean of every feature the model is trained on. A non-bloom day is represented by a 0, while a bloom day is 1. By looking at the table, we see clearly that when the water temperature is warmer, the blooms tend to be less frequent. This is the same for the Schmidt stability, more the stability is high less the blooms tend to happen.

From that analysis, we are quite confident about what the model says and how blooms are predicted during the year.

6.2 Limitations

During this research, we came across diverse limitations that lower the quality and the performance of our models.

The first one, and maybe the most notable one, is the size of the dataset, indeed, the dataset is made on a single-lake observations, so the diversity is therefore directly impacted, as is the number of observations made.

Afterwards, to be able to transpose the model between the two lakes, we needed to made some choices, in particular omit in-lake variables, and those choices result in the performance of the model, although the performance stays pretty high.

Finally, we cannot affirm that the predictions made for the Lake Neuchâtel are correct because we have no validation, due to a lack of in-lake data.

6.3 Future work

The next step would be to make in-lake observations and analysis in the Lake Neuchâtel and that way we can verify if the model is correct.

If the model turns out to be correct and the predictions correspond to the actual data and observations, this would mean that towns and villages bordering the lake could use this model to prevent cyanobacteria-related health problems among their citizens.

7. Conclusion

This research allowed us to predict high probability of bloom periods during the year by learning and training on a dataset from the Lake Greifensee.

The main result is that blooms generally appear during season of average and colder temperatures, principally in early spring and autumn.

We have to remember that it is not an absolute prediction of probabilities because the model is not trained on data from the Lake Neuchâtel but from a lake that has same environmental conditions. However, the model gives a good idea of the curve during the year even without the most precise variables.

References

- Raw Dataset, *EAWAG / Datalakes Greifensee* : https://www.datalakes-eawag.ch/datadetail/1387

- Weather API, *Open-Meteo* : https://open-meteo.com/

- Planktrothix in peri-alpine lakes, *EAWAG* : https://www.eawag.ch/en/info/portal/news/news-detail/cyanobacteria-in-lakes-risks-linked-to-loss-of-diversity/

Appendix

The classification report gives us a good idea of the whole performances and the metrics of our model.

	precision	recall	f1-score	support
0	0.92	0.83	0.87	379
1	0.60	0.79	0.68	127
accuracy			0.82	506
macro avg	0.76	0.81	0.78	506
weighted avg	0.84	0.82	0.82	506

Table 3 Classification report, Greifensee

The model catches 79% of the blooms (recall) with a precision of 0.60. For a warning use case, missing a bloom is worse than a false alarm, so favouring recall is the right call.

		Predicted	
		Positive	Negative
Actual	Positive	313 (True Positive)	66 (False Negative)
	Negative	27 (False Positive)	100 (True Negative)

Table 4 Confusion Matrix

The entire code of the models and analysis is open-source on a GitHub repository : https://github.com/segurad308-jpg/cyne-model-prediction